Path-Parameterization for Safe High Speed UAV Navigation

Lidia Rocha
Federal University of São Carlos and Venturus
Brazil
lidiarocha@ieee.org

Kelen Cristiane Teixeira Vivaldini
Federal University of São Carlos
Brazil
vivaldini@ufscar.br

***Abstract*— High-speed UAV navigation requires trajectory parameterization that jointly accounts for path geometry, obstacle clearance, and vehicle motion constraints. This work presents a three-stage framework combining B-spline geometric refinement, context-aware speed planning, and polynomial motion primitives. The main methodological contribution lies in the coupling of a local speed envelope based on curvature, obstacle proximity, and heading variation with a forward-backward feasibility pass, followed by transfer to dynamically feasible minimum-snap trajectory segments. The framework is evaluated through Software-in-the-Loop simulations using model predictive control in four environments with different geometric constraints. Comparisons with TOPPRA, MIQP, Motion Primitive Lattice Planning, and Cubic Polynomial Trajectory Generation examine path length, execution time, computational cost, acceleration, peak speed, and sustained high-speed operation. The results reveal scenario-dependent trade-offs between geometric efficiency, execution time, and dynamic characteristics rather than uniform superiority across all metrics.**

I. Introduction

High-speed Unmanned Aerial Vehicle (UAV) navigation introduces significant challenges to agility, responsiveness, and safety. These systems must rapidly adapt to dynamic environments while maintaining stable and precise flight behavior. Such capabilities are critical in mission-driven applications, including drone racing, emergency response, and autonomous delivery [1]. Achieving reliable high-speed flight requires more than accurate control, it depends on the coordinated integration of perception, planning, and execution [2].

A key requirement for reliable navigation at high speeds is path parameterization. While low-speed flight can be managed using simple, often discrete trajectories, high-speed operations demand smooth paths that ensure curvature continuity, gradual transitions, and compliance with the vehicle's dynamic constraints. Velocity prediction along a path is strongly affected by its geometric properties and contextual conditions [3], making geometry-aware optimization essential for maintaining dynamic feasibility [4]. Moreover, time-optimal waypoint tracking relies on adaptive speed profiles that respond to trajectory curvature and actuator limits [5].

Despite recent progress, existing methods still involve trade-offs. Energy-efficient trajectory planning methods can reduce mission costs but often rely on simplified speed assumptions that become unsafe at high speeds [6]. Other approaches based on reachability analysis [7], nonlinear model predictive control [8], or mixed-integer optimization [9] tend to be computationally expensive or lack the flexibility required for real-time deployment in dynamic environments.

This paper presents a complete path-parameterization framework tailored for high-speed UAVs operating in geometrically constrained and obstacle-rich environments. The proposed framework refines planned paths through three main modules. First, a geometric smoothing stage enforces curvature continuity while reducing jerk. Second, a curvature-aware speed planner adjusts the velocity profile based on local curvature and heading variations. Finally, a motion-primitive layer generates dynamically feasible trajectory segments for execution.

The main contributions of this work are summarized as follows:

- We formulate a context-aware velocity envelope that jointly accounts for local path curvature, obstacle clearance, and heading variation before enforcing longitudinal acceleration feasibility through a forward-backward pass.
- We couple geometric B-spline refinement and the resulting velocity parameterization with seventh-order minimum-snap motion primitives, preserving continuity up to jerk for trajectory execution.
- We evaluate the complete parameterization framework in Software-in-the-Loop simulation against four established trajectory-generation and path-parameterization approaches under different geometric constraints, and characterize its performance across environments with different geometric constraints.

The evaluated scenarios reveal different trade-offs between path length, execution time, acceleration, peak speed, and sustained high-speed operation. Rather than outperforming all baselines in every metric, the proposed framework aims to provide a balanced combination of geometric smoothness, dynamically feasible velocity allocation, and high-speed trajectory execution.

II. Related Works

Motion planning for UAVs in dynamic or uncertain environments has been extensively studied, with emphasis on improving flight efficiency, robustness, and real-time performance. Existing approaches include energy-aware speed estimation [10], computationally efficient motion primitives [11], uncertainty-aware planning through motion primitive sampling [12], and high-speed planning strategies that account for aerodynamic drag [13].

Trajectory refinement methods improve feasibility by reducing abrupt control commands while preserving safety and time efficiency. Representative approaches include optimization-based smoothing [14], learning-based trajectory refinement [15], and recursive smoothing techniques that reduce oscillations and improve tracking stability [16].

Speed planning has likewise received considerable attention, with methods based on spatio-temporal corridors [17], iterative refinement [18], quadratic optimization [19], trajectory optimization [20], and Gaussian-process-based refinement [21]. More recently, high-speed safety has been addressed through geofencing with control barrier functions [22] and safety-assured navigation frameworks [23]. However, these approaches typically emphasize either reactive safety or computational efficiency, rather than jointly addressing trajectory smoothness, adaptive speed planning, and high-speed dynamic feasibility.

The individual components employed in this work, including B-spline trajectory smoothing, forward-backward velocity parameterization, and minimum-snap polynomial trajectories, are well established in the literature. Therefore, the contribution of the proposed framework does not lie in introducing a new spline or polynomial representation. Instead, it lies in the coupling between geometric refinement and a context-dependent speed envelope that simultaneously considers curvature, obstacle clearance, and heading variation, and in propagating the resulting parameterization to the motion-primitive representation used for execution.

III. Problem Statement

We consider a UAV navigating from an initial position $\mathbf{p}_0$ to a goal position $\mathbf{p}_g$ in a bounded workspace $\mathcal{W} \subset \mathbb{R}^3$. The framework receives a reference trajectory generated by a global planner,

$$\mathcal{X} = \{(\mathbf{p}_i, \mathbf{v}_i)\}_{i=1}^{N}, \tag{1}$$

where $\mathbf{p}_i, \mathbf{v}_i \in \mathbb{R}^3$ denote the sampled position and velocity states. The objective is to transform this reference into a smooth and dynamically feasible trajectory

$$\mathbf{p}(t) = [x(t), y(t), z(t)]^\top, \qquad t \in [0, T], \tag{2}$$

while minimizing traversal time.

Assuming a point-mass model, the trajectory must satisfy

$$\mathbf{p}(0) = \mathbf{p}_0, \qquad \mathbf{p}(T) = \mathbf{p}_g, \tag{3}$$

$$\mathbf{p}(t) \in \mathcal{W}_{\text{free}}, \quad \|\dot{\mathbf{p}}(t)\| \le v_{\max}, \quad \|\ddot{\mathbf{p}}(t)\| \le a_{\max}, \tag{4}$$

for all $t \in [0, T]$, where $\mathcal{W}_{\text{free}} = \mathcal{W} \setminus \mathcal{O}$. The refined trajectory must also remain consistent with the reference samples,

$$\|\mathbf{p}(t_i) - \mathbf{p}_i\| \le \varepsilon, \qquad 0 < t_1 < \cdots < t_N < T. \tag{5}$$

The resulting trajectory is therefore collision-free, dynamically feasible, and suitable for high-speed execution while reducing high-order motion discontinuities.

IV. Smoothing

To ensure trajectory continuity and reduce higher-order motion discontinuities, we introduce a smoothing stage that transforms a discrete sequence of translational states into a continuous and differentiable trajectory suitable for high-speed execution. This optimization is performed once after the global path is generated.

The input trajectory is represented by the sequence

$$\mathcal{X} = \{(\mathbf{p}_i, \mathbf{v}_i)\}_{i=1}^{N}, \tag{6}$$

where

$$\mathbf{p}_i = [x_i, y_i, z_i]^\top, \qquad \mathbf{v}_i = [v_{x,i}, v_{y,i}, v_{z,i}]^\top \tag{7}$$

denote the position and velocity associated with the i-th sampled state. The states are generated at a constant sampling interval Δt, such that

$$t_i = (i-1)\Delta t, \qquad T = (N-1)\Delta t. \tag{8}$$

A time-parameterized B-spline trajectory

$$\mathbf{p}(t) : [0, T] \to \mathbb{R}^3 \tag{9}$$

is fitted to the sampled states and expressed as

$$\mathbf{p}(t) = \sum_{j=0}^{n} B_{j,d}(t)\mathbf{P}_j, \tag{10}$$

where $\mathbf{P}_j$ are the spline control points and $B_{j,d}(t)$ are the B-spline basis functions of degree $d \ge 3$.

The control points are obtained by minimizing a fitting objective that preserves consistency with the sampled positions and velocities,

$$J_{\text{fit}} = \sum_{i=1}^{N} \|\mathbf{p}(t_i) - \mathbf{p}_i\|^2 + \lambda_v \sum_{i=1}^{N} \|\dot{\mathbf{p}}(t_i) - \mathbf{v}_i\|^2. \tag{11}$$

Since the spline is parameterized with respect to time, its third derivative corresponds to the physical jerk of the UAV. Additional trajectory smoothness is promoted by minimizing

$$J_{\text{jerk}} = \int_0^T \|\dddot{\mathbf{p}}(t)\|^2 \, dt. \tag{12}$$

The smoothing stage, therefore, solves

$$\min_{\{\mathbf{P}_j\}} \; J_{\text{fit}} + \lambda_j J_{\text{jerk}}, \tag{13}$$

where λ_j controls the trade-off between trajectory fidelity and smoothness. The resulting trajectory preserves the original reference states while reducing jerk and improving suitability for high-speed flight.

In the implementation used in the experiments, the geometric refinement is instantiated as a cubic B-spline ($d = 3$). The spline is evaluated using an interpolating configuration with smoothing parameter $s = 0$, and the resulting curve is uniformly resampled using 600 points. Consequently,

no additional empirical smoothing coefficient is introduced during the spline interpolation stage. These parameters are kept fixed for all evaluated scenarios.

V. Speed Planning

Speed planning ensures that the UAV adjusts its velocity along the trajectory to remain within dynamic limits while preserving time efficiency. The velocity profile is adapted based on three factors: local curvature, proximity to obstacles, and heading changes. This strategy prevents aggressive acceleration or braking in regions where the vehicle's stability or responsiveness may be compromised.

Let $\kappa(s)$ denote the curvature at arc-length parameter s. The maximum allowable speed $v(s)$ is computed as:

$$v(s) = \min\left\{v_{\max}, v_{\text{curv}}(s), v_{\text{obs}}(s), v_{\text{yaw}}(s)\right\}, \tag{14}$$

where

$$v_{\text{curv}}(s) = \sqrt{\frac{a_{\text{lat}}^{\max}}{\max\left(\kappa(s), \varepsilon\right)}}, \tag{15}$$

$$v_{\text{obs}}(s) = f\left(d_{\text{min}}(s)\right), \tag{16}$$

$$v_{\text{yaw}}(s) = g\left(\Delta\psi(s)\right). \tag{17}$$

Here, $a_{\text{lat}}^{\max}$ is the maximum allowable lateral acceleration, $d_{\text{min}}(s)$ is the minimum distance to nearby obstacles, and $\Delta\psi(s)$ is heading variation between consecutive trajectory segments. The small positive constant ε avoids numerical singularities in straight-line segments where $\kappa(s) = 0$.

The obstacle-dependent velocity function $f(\cdot)$ is implemented as

$$v_{\text{obs}}(s) = v_{\min} + (v_{\max} - v_{\min})\,\text{clip}\left(\frac{d_{\min}(s) - d_{\text{safe}}}{d_{\text{slow}} - d_{\text{safe}}}, 0, 1\right), \tag{18}$$

where $d_{\text{safe}} = 0.35$ m denotes the minimum obstacle clearance and $d_{\text{slow}} = 2.5$ m is the distance above which obstacle proximity no longer restricts the reference velocity.

The heading-dependent velocity function $g(\cdot)$ is defined as

$$v_{\text{yaw}}(s) = \text{clip}\left(\frac{v_{\max}}{1 + k_\psi |\Delta\psi(s)|}, v_{\min}, v_{\max}\right), \tag{19}$$

where $k_\psi = 2.2$ controls the reduction in reference speed caused by local heading variation. The heading profile is obtained from the local path derivatives as

$$\psi(s) = \text{atan2}\left(\frac{dy}{ds}, \frac{dx}{ds}\right). \tag{20}$$

For all experiments, the speed-planning parameters are fixed to $v_{\min} = 0.5$ m/s, $v_{\max} = 12.0$ m/s, $a_{\max} = 1.7$ m/s^2, and $a_{\text{lat}}^{\max} = 2.0$ m/s^2.

A forward-backward pass is applied to generate a feasible speed profile. The forward pass respects acceleration bounds:

$$v_f(s_{i+1}) = \min\left\{v(s_{i+1}),\ \sqrt{v_f^2(s_i) + 2a_{\max}\Delta s}\right\}, \tag{21}$$

and the backward pass guarantees deceleration feasibility:

$$v_b(s_i) = \min\left\{v_f(s_i),\ \sqrt{v_b^2(s_{i+1}) + 2a_{\max}\Delta s}\right\}. \tag{22}$$

The final profile is defined as $v^*(s_i) = \min\left\{v_f(s_i), v_b(s_i)\right\}$, ensuring consistency with acceleration limits in both directions.

The final feasible velocity profile is filtered using a five-sample moving-average window and clipped to the interval $[v_{\min}, v_{\max}]$. The initial and terminal velocities are subsequently fixed to $v_{\min}$.

Uniform speed execution can lead to unsafe behavior at high speeds due to reduced reaction time, overshooting in curves, and insufficient braking distance. The proposed curvature-aware speed adaptation mitigates these issues by modulating velocity based on geometric and contextual factors, improving tracking stability and maintaining safety margins.

VI. Motion Primitives

After trajectory smoothing and speed planning, the time-parameterized trajectory is represented as a sequence of polynomial motion primitives for controller execution. Each primitive is defined as

$$\mathbf{p}(t) = \sum_{i=0}^{7} \mathbf{a}_i t^i, \qquad t \in [0, T_p], \tag{23}$$

where $\mathbf{a}_i \in \mathbb{R}^3$ are the polynomial coefficients and T_p is the primitive duration. Seventh-order polynomials provide sufficient degrees of freedom to enforce boundary conditions up to jerk while minimizing snap,

$$\min_{\mathbf{p}(t)} \int_0^{T_p} \left\| \frac{d^4\mathbf{p}(t)}{dt^4} \right\|^2 dt, \tag{24}$$

subject to

$$\mathbf{p}^{(k)}(0) = \mathbf{b}_0^{(k)}, \qquad \mathbf{p}^{(k)}(T_p) = \mathbf{b}_T^{(k)}, \qquad k = 0, 1, 2, 3. \tag{25}$$

The smoothed trajectory is partitioned into consecutive intervals, and one primitive is generated from the corresponding position, velocity, acceleration, and jerk boundary states. This preserves the planned velocity profile and continuity while providing a compact dynamically feasible representation for direct UAV execution.

VII. Results

The framework is evaluated in simulation using path length, execution time, trajectory-generation time, average acceleration, maximum speed, and the percentage of execution time above 70% and 90% of the maximum observed speed. Path length and average acceleration are computed as

$$L = \int_0^T \|\dot{\mathbf{p}}(t)\| dt, \qquad \bar{a} = \frac{1}{T}\int_0^T \|\ddot{\mathbf{p}}(t)\| dt, \tag{26}$$

where T is the trajectory duration. The high-speed ratio is

$$P_\alpha = \frac{100}{T}\int_0^T \mathbb{I}\left(\|\dot{\mathbf{p}}(t)\| \geq \alpha v_{\max}\right) dt, \tag{27}$$

where $\alpha \in \{0.7, 0.9\}$ and $v_{\max}$ is the maximum observed speed of the corresponding execution. Generation time denotes the wall-clock time required to generate the trajectory.

Four synthetic environments are considered: *Spiral Confinement* (15×15 m), with sustained curvature; *Bottleneck* (20×10 m), with narrow passages; *Zigzag Channel* (12×6 m), with repeated heading changes; and *Open Field with Obstacles* (25×25 m), with sparse obstacles.

The proposed framework is compared with TOPPRA [7], MIQP [24], Motion Primitive Lattice Planning [25], and Cubic Polynomial Trajectory Generation [26]. Baseline parameters are fixed across all scenarios. MIQP uses 250 discretization points and objective weights of 20, 5, and 50 for reference tracking, first-order variation, and second-order smoothness, respectively, and is implemented in CVXPY using OSQP, with ECOS/SCS as fallbacks. Motion Primitive Lattice Planning uses 1.0 m primitives and 15° heading discretization. Cubic Polynomial uses natural cubic interpolation with 300 samples. TOPPRA uses per-axis velocity and acceleration limits of ± 8 m/s and ± 1.7 m/s^2, respectively, with `ParametrizeConstAccel` and 100 temporal samples.

MIQP, Motion Primitive, and Cubic Polynomial trajectories use the same downstream speed-parameterization and motion-execution pipeline, whereas TOPPRA retains its native time parameterization. This setup isolates geometric differences while maintaining consistent execution conditions.

Simulations are performed on a 13th-generation Intel Core i7 platform with 64 GB RAM using an F450 quadrotor model and MPC trajectory tracking. The vehicle mass is 3.5 kg. Dynamic limits include a 0.05 s minimum control interval, thrust limits of 5–25 m/s^2, and a maximum rotational velocity of 25 rad/s. Environment occupancy is represented using OctoMap [27].

All environments and algorithm configurations are deterministic; therefore, each reported value corresponds to a single execution rather than a statistical average. Bold values indicate the numerically best result for each metric and do not imply statistical significance.

A. Baseline Comparisons

Figures 1–3 summarize the geometric and dynamic behavior of the evaluated methods. The proposed framework produces smooth transitions between successive path segments, while velocity reductions occur mainly near regions of stronger curvature, heading changes, and reduced obstacle clearance. The resulting position, velocity, and acceleration profiles show continuous velocity evolution, with acceleration changes concentrated around trajectory transitions.

Table I summarizes the Spiral Confinement results. The proposed method achieves the shortest execution time (21.80 s) and the largest fraction above the 90% speed threshold (34.81%). Motion Primitive generates the shortest path, while MIQP reaches the highest peak speed and the largest fraction above 70%. Cubic Polynomial presents the lowest average acceleration. These results highlight a trade-off between geometric efficiency, dynamic smoothness, and sustained high-speed operation.

Table II presents the Bottleneck results. The proposed method produces the numerically shortest path (18.59 m), although the 0.01–0.03 m differences relative to MIQP, Cubic Polynomial, and TOPPRA are considered negligible. MIQP and Cubic Polynomial achieve the shortest execution time and lowest average acceleration, whereas the proposed method maintains the largest fractions above both high-speed thresholds. TOPPRA reaches the highest peak speed, while Motion Primitive provides the lowest generation time.

Table III reports the Zigzag results. Motion Primitive produces the shortest path and the largest fraction above the 90% threshold. MIQP and Cubic Polynomial achieve the shortest execution time and lowest average acceleration. The proposed method, however, reaches the highest peak speed and maintains 96.61% of the execution above the 70% threshold, illustrating the trade-off between path length and sustained velocity under repeated heading changes.

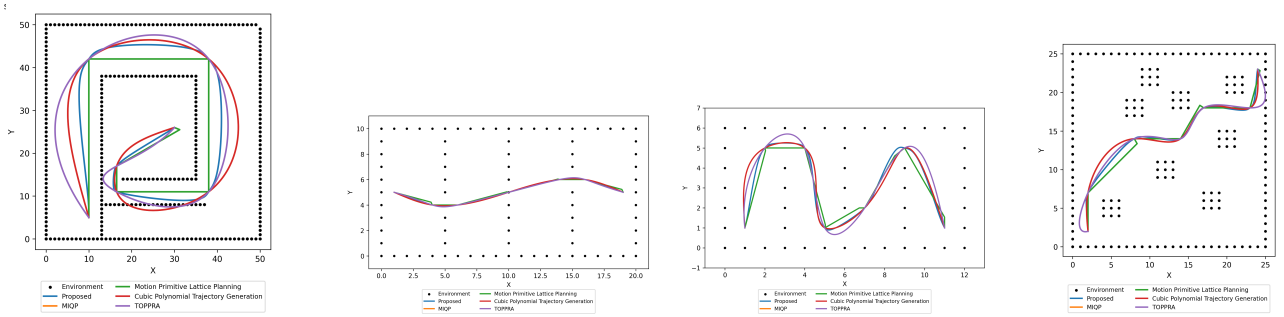
Table IV summarizes the Open Field results. The proposed method achieves the shortest path, lowest average acceleration, and highest peak speed. Motion Primitive provides the shortest execution and generation times, whereas MIQP achieves the largest fractions above both high-speed thresholds. Cubic Polynomial shows similar sustained-speed behavior to MIQP.

The results show scenario-dependent trade-offs rather than uniform dominance by a single method. The proposed framework provides a balanced combination of geometric efficiency, adaptive velocity allocation, and competitive execution performance.

VIII. Conclusion

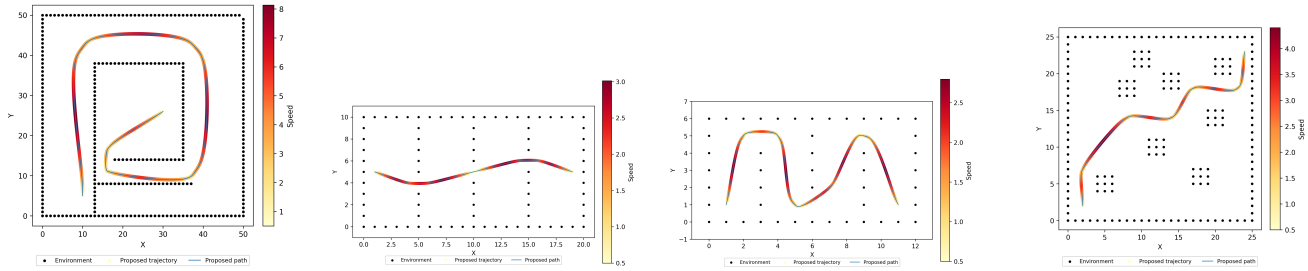
This work presented a path-parameterization framework for high-speed UAV navigation combining B-spline trajectory refinement, context-aware speed planning, and polynomial motion primitives. The main contribution lies in coupling trajectory geometry with a local velocity envelope based on curvature, obstacle proximity, and heading variation, followed by acceleration-feasible velocity propagation and dynamically feasible polynomial execution.

Software-in-the-Loop experiments across four geometrically distinct environments show that the framework provides different advantages depending on the scenario, including sustained high-speed operation, low acceleration, short geometric paths, or competitive execution time. The results do not indicate uniform dominance over all baselines and metrics, but rather scenario-dependent trade-offs between geometric efficiency and temporal performance. Future work



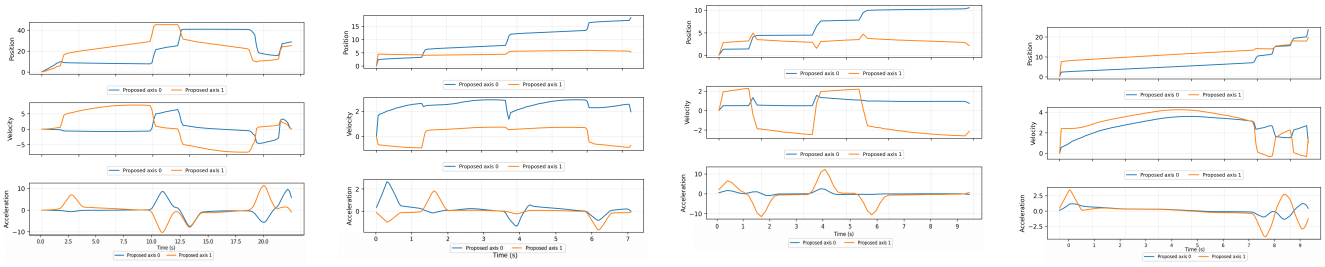
((a)) Spiral ((b)) Bottleneck ((c)) ZigZag ((d)) Open Field

Fig. 1: Comparison between the proposed method and the baselines across different scenarios. X and Y axes in meters.



((a)) Spiral ((b)) Bottleneck ((c)) ZigZag ((d)) Open Field

Fig. 2: Proposed algorithm trajectory with speed profile across different scenarios. X and Y axes in meters.



((a)) Spiral ((b)) Bottleneck ((c)) ZigZag ((d)) Open Field

Fig. 3: Position, velocity, and acceleration profiles of the proposed algorithm across different scenarios. Time (s) vs position (m), velocity (m/s), and acceleration (m/s²).

TABLE I: Performance comparison in the Spiral Confinement scenario.

Algorithm	Path Length (m)	Exec. Time (s)	Gen. Time (s)	Avg. Acc. (m/s^2)	Max. Speed (m/s)	Time >70% (%)	Time >90% (%)
Proposed	144.63	**21.80**	0.00094	1.313	7.85	64.44	**34.81**
MIQP	151.00	22.40	0.00063	0.326	**8.79**	**77.80**	19.00
Motion Primitive	**141.82**	23.80	**0.00035**	1.222	7.75	58.62	13.10
Cubic Polynomial	151.00	22.70	0.00124	**0.285**	8.76	75.94	30.34
TOPPRA	153.59	38.50	1.50546	1.082	5.22	63.93	29.86

TABLE II: Performance comparison in the Bottleneck scenario.

Algorithm	Path Length (m)	Exec. Time (s)	Gen. Time (s)	Avg. Acc. (m/s^2)	Max. Speed (m/s)	Time >70% (%)	Time >90% (%)
Proposed	**18.59**	7.20	0.00093	0.402	3.03	**94.48**	**46.21**
MIQP	18.60	**6.80**	0.00060	**0.310**	3.29	60.00	15.00
Motion Primitive	18.77	8.60	**0.00010**	1.964	2.91	45.45	31.82
Cubic Polynomial	18.60	**6.80**	0.00121	**0.310**	3.30	58.00	14.00
TOPPRA	18.62	8.90	5.63943	1.083	**3.71**	22.04	7.62

TABLE III: Performance comparison in the Zigzag Channel scenario.

Algorithm	Path Length (m)	Exec. Time (s)	Gen. Time (s)	Avg. Acc. (m/s^2)	Max. Speed (m/s)	Time >70% (%)	Time >90% (%)
Proposed	21.24	9.20	0.00093	0.886	**2.83**	**96.61**	23.73
MIQP	21.29	**9.00**	0.00061	**0.600**	2.68	80.00	25.00
Motion Primitive	**20.73**	16.00	**0.00010**	0.875	1.62	51.92	**30.77**
Cubic Polynomial	21.29	**9.00**	0.00114	**0.600**	2.69	79.00	24.00
TOPPRA	22.13	13.00	1.76811	1.019	2.36	29.26	6.61

TABLE IV: Performance comparison in the Open Field with Obstacles scenario.

Algorithm	Path Length (m)	Exec. Time (s)	Gen. Time (s)	Avg. Acc. (m/s^2)	Max. Speed (m/s)	Time >70% (%)	Time >90% (%)
Proposed	**36.88**	9.30	0.00089	**0.588**	**5.58**	61.59	33.11
MIQP	37.40	9.00	0.00065	0.819	5.40	**81.01**	**58.23**
Motion Primitive	37.67	**8.70**	**0.00013**	1.855	4.88	79.00	40.00
Cubic Polynomial	37.40	9.10	0.00125	0.852	5.38	80.00	57.50
TOPPRA	39.19	14.50	1.77590	1.148	3.87	17.64	4.61

will extend the evaluation to randomized environment configurations, dynamic obstacles, and physical UAV experiments.

ACKNOWLEDGMENT

The authors would like to thank the FAPESP for grant 2025/16918-4, the CNPq for grant 420753/2025-7, and the CAPES for their financial support.

REFERENCES

[1] L. Rocha, M. Saska, and K. Vivaldini, "Overview of uav trajectory planning for high-speed flight," in *2023 International Conference on Unmanned Aircraft Systems (ICUAS)*. IEEE, 2023, pp. 110–117.

[2] D. Hanover, A. Loquercio, L. Bauersfeld, A. Romero, R. Penicka, Y. Song, G. Cioffi, E. Kaufmann, and D. Scaramuzza, "Autonomous drone racing: A survey," *IEEE Transactions on Robotics*, 2024.

[3] H. Yang, C. Liu, M. Zhu, X. Ban, and Y. Wang, "How fast you will drive? predicting speed of customized paths by deep neural network," *IEEE transactions on intelligent transportation systems*, vol. 23, no. 3, pp. 2045–2055, 2021.

[4] Z. Wang, X. Zhou, C. Xu, and F. Gao, "Geometrically constrained trajectory optimization for multicopters," *IEEE Transactions on Robotics*, vol. 38, no. 5, pp. 3259–3278, 2022.

[5] P. Foehn, A. Romero, and D. Scaramuzza, "Time-optimal planning for quadrotor waypoint flight," *Science robotics*, vol. 6, no. 56, p. eabh1221, 2021.

[6] D. Datsko, F. Nekovar, R. Penicka, and M. Saska, "Energy-aware multi-uav coverage mission planning with optimal speed of flight," *IEEE Robotics and Automation Letters*, vol. 9, no. 3, pp. 2893–2900, 2024.

[7] H. Pham and Q.-C. Pham, "A new approach to time-optimal path parameterization based on reachability analysis," *IEEE Transactions on Robotics*, vol. 34, no. 3, pp. 645–659, 2018.

[8] Y. Du, J. Kang, X. Tian, M. Wang, D. Liu, and M. Cong, "Nonlinear mpc of quadrotors for path planning and dynamic obstacle avoidance," in *2024 IEEE 14th International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*. IEEE, 2024, pp. 485–490.

[9] D. Mellinger, A. Kushleyev, and V. Kumar, "Mixed-integer quadratic program trajectory generation for heterogeneous quadrotor teams," in *2012 IEEE international conference on robotics and automation*. IEEE, 2012, pp. 477–483.

[10] L. Bauersfeld and D. Scaramuzza, "Range, endurance, and optimal speed estimates for multicopters," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2953–2960, 2022.

[11] M. W. Mueller, M. Hehn, and R. D'Andrea, "A computationally efficient motion primitive for quadrocopter trajectory generation," *IEEE transactions on robotics*, vol. 31, no. 6, pp. 1294–1310, 2015.

[12] G. M. Gutow, J. D. Rogers, and K. Williams, "Motion primitive path planning under uncertainty for high-speed vehicles," in *AIAA SCITECH 2023 Forum*, 2023, p. 1258.

[13] C. Toumieh and A. Lambert, "High-speed planning in unknown environments for multirotors considering drag," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2021, pp. 7844–7850.

[14] B. Zhou, F. Gao, L. Wang, C. Liu, and S. Shen, "Robust and efficient quadrotor trajectory generation for fast autonomous flight," *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 3529–3536, 2019.

[15] S. Meraglia and M. Lovera, "Smoother-based iterative learning control for uav trajectory tracking," *IEEE Control Systems Letters*, vol. 6, pp. 1501–1506, 2021.

[16] N. A. Letizia, B. Salamat, and A. M. Tonello, "A novel recursive smooth trajectory generation method for unmanned vehicles," *IEEE Transactions on Robotics*, vol. 37, no. 5, pp. 1792–1805, 2021.

[17] W. Ding, L. Zhang, J. Chen, and S. Shen, "Safe trajectory generation for complex urban environments using spatio-temporal semantic corridor," *IEEE Robotics and Automation Letters*, vol. 4, no. 3, pp. 2997–3004, 2019.

[18] K. Okumura, Y. Tamura, and X. Défago, "Iterative refinement for real-time multi-robot path planning," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2021, pp. 9690–9697.

[19] Y. Zhang, H. Sun, J. Zhou, J. Pan, J. Hu, and J. Miao, "Optimal vehicle path planning using quadratic optimization for baidu apollo open platform," in *2020 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2020, pp. 978–984.

[20] W. Xu, J. Wei, J. M. Dolan, H. Zhao, and H. Zha, "A real-time motion planner with trajectory optimization for autonomous vehicles," in *2012 IEEE international conference on robotics and automation*. IEEE, 2012, pp. 2061–2067.

[21] J. Cheng, Y. Chen, Q. Zhang, L. Gan, C. Liu, and M. Liu, "Real-time trajectory planning for autonomous driving with gaussian process and incremental refinement," in *2022 International Conference on Robotics and Automation (ICRA)*. IEEE, 2022, pp. 8999–9005.

[22] A. Singletary, A. Swann, Y. Chen, and A. D. Ames, "Onboard safety guarantees for racing drones: High-speed geofencing with control barrier functions," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2897–2904, 2022.

[23] Y. Ren, F. Zhu, G. Lu, Y. Cai, L. Yin, F. Kong, J. Lin, N. Chen, and F. Zhang, "Safety-assured high-speed navigation for mavs," *Science Robotics*, vol. 10, no. 98, p. eado6187, 2025.

[24] R. Quirynen and S. Di Cairano, "Sequential quadratic programming algorithm for real-time mixed-integer nonlinear mpc," in *2021 60th IEEE Conference on Decision and Control (CDC)*. IEEE, 2021, pp. 993–999.

[25] M. Tiger, D. Bergström, A. Norrstig, and F. Heintz, "Enhancing lattice-based motion planning with introspective learning and reasoning," *IEEE Robotics and Automation Letters*, 2021.

[26] H. Hong, R. Tekin, and F. Holzapfel, "Guaranteed smooth trajectory generation for field-of-view constrained impact-time control," *Journal of Guidance, Control, and Dynamics*, 2021.

[27] A. Hornung, K. M. Wurm, M. Bennewitz, C. Stachniss, and W. Burgard, "Octomap: An efficient probabilistic 3d mapping framework based on octrees," *Autonomous robots*, vol. 34, pp. 189–206, 2013.